



COMSATS University Islamabad

Department of Computer Science

Course Syllabus

Course Information

Course Code: AIC372

Credit Hours: 3(3,0)

Lab Hours/Week: 0

Course Title: **Knowledge Representation and Reasoning**

Lecture Hours/Week: 3

Pre-Requisites: AIC262

Catalogue Description:

The course introduces the principles of logic-based knowledge representation and reasoning. Topics include: Overview; Knowledge Representations; Inheritance & Defaults; Temporal Relations; Ontologies & Commonsense Knowledge; Semantic Web; Resolution; Horn Clauses; Reasoning Systems for Categories, Reasoning with Default Information; and Procedural Control of Reasoning.

Reading Material:

Text Book:

1. Knowledge representation and reasoning, Blokdyk, G., 5STARCook, 2020.

Reference Books:

1. Knowledge Representation, Reasoning, and the Design of Intelligent Agents: The Answer-Set Programming Approach, Gelfond, M., Kahl, Y., Cambridge University Press, 2014.
2. Knowledge Representation and Reasoning, Brachman, R., Levesque, H., The Morgan Kaufmann Series in Artificial Intelligence, 2004.
3. Knowledge Representation, Reasoning, and the Design of Intelligent Agents, by Michael Gelfond and Yulia Kahl.

Week wise Plan:

Lecture #	CDF Unit #	Topics Covered	Reference Material
1.	1	Key Concepts: Knowledge, Representation, Reasoning. Why Knowledge Representation and Reasoning?.	Levesque: Ch1
2.	1	Knowledge-Based Systems, Why Knowledge Representation? Why Reasoning?, The Role of Logic, Relationship between Knowledge and Artificial Intelligence.	Levesque: Ch1
3.	2	Knowledge Engineering, Vocabulary, Facts: Complex, Terminological, Entailments, Other Facts.	Levesque: Ch3
4.	2	Rules in Production: Basic operations, working memory, Examples, Conflict Resolution, Enhanced Production Systems, Applications and Advantages, Significant Production Rule Systems.	Levesque: Ch7
5.	2	Object-Oriented Representation: Objects and Frames, Frame formalism; Generic, Inheritance and Reasoning with Frames, Example.	Levesque: Ch8
6.	2	Object-Oriented Representation: Uses of frames, Extensions to the Frame Formalism, Object-Driven Programming with Frames.	Levesque: Ch8
7.	2	Inheritance: Inheritance Networks; Strict Inheritance, Defeasible Inheritance. Strategies for Defeasible Inheritance: Shortest Path Heuristics, Limitations of Shortest Path, Inferential Distance.	Levesque: Ch10

		Formal Account of Inheritance Network; Extensions, Inheritance Reasoning.	
8.	2	Defaults: Introduction: Generics and Universals, Default Reasoning, Nonmonotonicity; Closed-World Reasoning: Assumption, Completeness of Knowledge, Query Evaluation, Quantifiers and Domain Closure. Circumscription: Minimal Entailment, Fixed and Variable Predicates. Default Logic: Rules, Extensions (default, multiple).	Levesque: Ch11
9.	2	Defaults: Autoepistemic Logic, Conclusion Uncertainty: Noncategorical Reasoning, Objective Probability, Subjective Probability. Vagueness: Conjunction and Disjunction, Rules, and Bayesian Reconstruction.	Levesque: Ch11, Ch12
10.	3	First-Order Logic (FOL): Syntax, Semantics; Interpretations, Denotation, Satisfaction. Pragmatics: Logical Consequence, Why to care?	Levesque: Ch2
11.	3	Explicit and Implicit Belief (FOL): Example, Knowledge-based Systems. The Promotional Case: Resolution derivations, Entailment Procedure.	Levesque: Ch2, Ch4
12.	3	FOL: Handling Variables and Quantifiers, First-order Resolution.	Levesque: Ch4
13.	3	FOL: Answer Extraction, Skolemization, Equality.	Levesque: Ch4
14.	3	Dealing with Computational Intractability: First-Order Logic case, Herbrand Theorem.	Levesque: Ch4
15.	3	Propositional Case, Implications, SAT solvers, General unifiers, Other refinements. Computing SLD derivations: Reasoning; First Order Case.	Levesque: Ch4, Ch5
16.	Mid Term Exam		
17.			
18.	4	Procedural Control of Reasoning: Facts and rules, Rule formation and Search Strategy, Algorithm Design, Specifying Goal Order.	Levesque: Ch6
19.	4	Committing to Proof Methods, Controlling Backtracking, Negation as Failure, Dynamic Databases, Production System and Case Study.	Levesque: Ch6, Ch7
20.	4	Objective Probability: Basic Postulates, Conditional Probability and Independence; Subjective Probability: Statistics of Belief.	Levesque: Ch12
21.	4	Basic Bayesian Approach, Belief Networks, Example of Network, Influence diagram, Dempster-Shafer Theory.	Levesque: Ch12
22.	4	Tradeoff between Representation and Reasoning: Description Logic Case study; Two Logic Languages, Computing Subsumption. Limited Languages, What makes reasoning hard?	Levesque: Ch16
23.	4	Vivid Knowledge; Analogues, Diagrams, Models. Beyond Vivid; Sets of Literals, Incorporating Definitions, Hybrid Reasoning.	Levesque: Ch16
24.	5	Horn Clauses; Resolution Derivations with Horn clauses. SLD Resolution; Goal Trees, Specifying Goal Order.	Levesque: Ch5, Ch6
25.	5	Computing SLD derivations; Backward Chaining, Controlling Backtracking, Forward Chaining, The First Order Case.	Levesque: Ch5, Ch6
26.	5	Answer Set Semantics: Interpretations, Positive programs, General Programs.	Chapters 2 and 6
27.	5	Answer Set Programming, Computing Hamiltonian Paths using ASP.	Chapters 2 and 6
28.	5	Answer Set Solvers; DLV, SMOLEDS, CLASP, Answer Set Planning.	
29.	6	Planning in Situation Calculus; Example, Using Resolution.	Levesque: Ch15

30.	6	STRIP Representation: Progressive Planning, Regressive Planning.	Levesque: Ch15
31.	6	Planning as a Reasoning Task: Avoiding Redundant Search, Application-dependent Control.	Levesque: Ch15
32.	6	Beyond Basics: Hierarchical Planning, Conditional Planning, “Even the Bes-laid Plans”.	Levesque: Ch15
Final Term Exam			

Student Outcomes (SOs)

S.#	Description
1	Understanding different types of Knowledge, types and methods of Representation. Students should be able to make reasoning over represented knowledge by understanding Knowledge Representation.
2	Creating by building context-specific knowledge representation system for given scenario/problem. Students should develop reasoning based Intelligent Systems.
3	By applying deductive systems using FOL, students should be able to demonstrate them.

Course Learning Outcomes (CLOs)

Sr.#	Unit #	Course Learning Outcomes	Blooms Taxonomy Learning Level	SO
CLO's for Theory				
CLO-1	1	Explain types of knowledge and their relation to intelligence.	<i>Understanding</i>	1
CLO-2	2	Build a context-specific knowledge representation system for a given problem.	<i>Creating</i>	2-3
CLO-3	3	Describe the concepts of variables, conditional, repetitive structures, and functions using a programming language.	<i>Applying</i>	1,2
CLO-4	4-6	Develop reasoning-based intelligent system	<i>Creating</i>	2-4

CLO Assessment Mechanism

Assessment Tools	CLO-1	CLO-2	CLO-3	CLO-4
Quizzes	Quiz 1	Quiz 2	Quiz 3	Quiz 4
Assignments	-	Assignment 1	Assignment 2	Assignment 3&4
Mid Term Exam	Mid Term Exam			-
Final Term Exam	Final Term Exam			

Policy & Procedures

- **Attendance Policy:** Every student must attend 80% of the lectures as well as participate in class activities in this course. The students falling short of required percentage of attendance of lectures, is not allowed to appear in the terminal examination.

- **Course Assessment:**

	Quizzes	Assignments	Mid Term Exam	Terminal Exam	Total
Theory(T)	15	10	25	50	100

- **Grading Policy:** The minimum passing marks for each course is 50% (In case of LAB; in addition to

theory, student is also required to obtain 50% marks in the lab to pass the course). The correspondence between letter grades, credit points, and percentage marks at CUI is as follows:

Grade	A	A-	B+	B	B-	C+	C	C-	D+	D	F
Marks	>= 85	80 - 84	75 - 79	71 - 74	68 - 70	64 - 67	61 - 63	58 - 60	54 - 57	50-53	< 50
Cr. Point	3.67-4.00	3.34-3.66	3.01-3.33	2.67-3.00	2.34-2.66	2.01-2.33	1.67-2.00	1.31-1.66	1.01-1.30	0.10-1.00	0.00

- **Missing Exam:** No makeup exam will be given for final exam under any circumstance. When a student misses the mid-term exam for a legitimate reason (such as medical emergencies), his grade for this exam will be determined based on the Department policy. Further, the student must provide an official excuse within one week of the missed exam.
- **Academic Integrity:** All CUI policies regarding ethics apply to this course. The students are advised to discuss their grievances/problems with their counsellors or course instructor in a respectful manner.
- **Plagiarism Policy:** Plagiarism, copying and any other dishonest behaviour is prohibited by the rules and regulations of CUI. Violators will face serious consequences.